5 **(a)** State what is meant by the mass defect of a nucleus.

.....

.....

.....[2]

(b) A stationary nucleus of uranium-238 ($^{238}_{92}U$) undergoes alpha decay to produce an excited nucleus of thorium-234 ($^{234}_{90}Th^{*}$). The kinetic energy of the emitted alpha particle is 4.151 MeV. A gamma-ray photon is then immediately emitted from the excited thorium nucleus after the alpha decay.

Assume that the rebound kinetic energy of the thorium nucleus is negligible.

Fig. 5.1 shows the masses of some of the nuclides involved in the decay reaction.

nuclide	nuclide mass/ u
$^{4}_{2}\alpha$	4.000407
$^{234}_{90}Th$	233.915174

Fig. 5.1

The total energy released in the decay of the nucleus of uranium-238 is 4.274 MeV.

(i) Calculate the mass, in u, of the uranium-238 nuclide.

mass = ... u [3]

(ii) Determine a value for the wavelength of the gamma radiation emitted during the decay of the uranium-238 nucleus.

wavelength = ... m [3]

(iii) In practice, the rebound kinetic energy of the thorium nucleus is **not** negligible.

Explain, without further calculation, how your answer in **(b)(ii)** compares with the true wavelength of gamma radiation emitted during the decay of the uranium-238 nucleus.

.....

.....[1]

(c) Gamma radiation emitted during the decay of a sample of uranium-238 has a single wavelength.

Nuclei of cobalt-60 ($^{60}_{27}Co$) decay by beta emission and also emit gamma radiation in the process.

Suggest why there is **not** a single wavelength for the gamma radiation emitted during the decay of a sample of cobalt-60.

.....

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.....

.....[2]

[Total: 11]